CHAPTER 12

LINEAR PROGRAMMING

CHAPTER ESSENCE AT A GLANCE:
Linear programming formulation, graphical feasible region method, corner point theorem, and finding optimal objective solutions under constraints.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Formulation of Linear Optimization:** Linear programming seeks to maximize or minimize a linear objective function subject to a system of simultaneous linear inequality constraints representing real-world resource bounds.

• **Convex Feasible Regions:** The simultaneous solution of all constraint half-planes creates a planar polygon known as the feasible region, which is always convex when non-empty.

• **The Corner Point Theorem:** Optimal extreme values of the linear objective function are guaranteed to occur at one or more vertex corner points of the bounded feasible region.

• **Unbounded Region Verification:** For unbounded feasible regions, a proposed optimum value is valid if and only if the open half-plane determined by the objective function shares no points with the feasible region.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Optimize } Z = ax + by \quad \text{subject to } A\mathbf{x} \leq \mathbf{b},\ x \geq 0,\ y \geq 0$$

Standard mathematical formulation of Linear Programming Problem

$$Z_{\max} = \max_{(x_i\,,\,y_i)\,\in\,\mathcal{V}} (ax_i + by_i), \quad Z_{\min} = \min_{(x_i\,,\,y_i)\,\in\,\mathcal{V}} (ax_i + by_i)$$

Corner Point Theorem evaluation at feasible region vertices

$$ax + by > M \quad \text{has no common points with feasible region} \rightarrow Z_{\max} = M$$

Optimality test for unbounded feasible region

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Feasible Region Shading] Always test origin (0, 0) in each inequality to decide correct half-plane! Shading the wrong side forfeits 3 marks.

• **Alert:** [Trap Alert: Corner Point Coordinates] Solve simultaneous equations of intersecting boundary lines algebraically to get exact fraction/integer coordinates for corner points.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define: (a) Objective function, (b) Constraints, (c) Feasible region, (d) Optimal solution in LPP. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[LPP Definitions]** 1. Objective Function: Linear function Z = ax + by of decision variables x and y that needs to be optimized (maximized or minimized). 2. Constraints: Linear inequalities or equations on decision variables imposed by resource limitations (e.g. x + y ≤ 4, x ≥ 0, y ≥ 0). 3. Feasible Region: The common region determined by all the constraints simultaneously. 4. Optimal Solution: Any point in the feasible region that gives the optimal (maximum or minimum) value of the objective function. [2 Marks]

• **[Boxed Result: Objective function (Z = ax + by), Constraints (inequalities), Feasible Region, Optimal Solution]**

2. Explain the Corner Point Method for solving a Linear Programming Problem. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Corner Point Method Algorithm]** 1. Formulate mathematical model of LPP. 2. Graph all linear inequalities and determine the common Feasible Region. 3. Find the coordinates of all extreme vertices (Corner Points) of the feasible region by solving intersection equations. 4. Evaluate the objective function Z = ax + by at each corner point. 5. For a bounded region: The highest value is the Maximum and lowest is the Minimum. 6. For an unbounded region: If M is candidate maximum, check whether open half-plane ax + by > M shares common points with feasible region. If no common points, M is maximum. [2 Marks]

• **[Boxed Result: Optimal value always occurs at one of the Corner Points of feasible region]**

3. Solve graphically: Maximize Z = 3x + 4y subject to x + y <= 4, x >= 0, y >= 0. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Constraints & Boundary Line]** Maximize Z = 3x + 4y subject to x + y ≤ 4, x ≥ 0, y ≥ 0. Boundary line x + y = 4 meets axes at (4, 0) and (0, 4). Test origin (0, 0): 0 + 0 ≤ 4 (True). Feasible region is triangle enclosed by (0,0), (4,0), (0,4) in first quadrant. [1.5 Marks]

• **[Corner Point Evaluation]** Corner Point (0, 0): Z = 3(0) + 4(0) = 0. Corner Point (4, 0): Z = 3(4) + 4(0) = 12. Corner Point (0, 4): Z = 3(0) + 4(4) = 16 (Maximum). [1.5 Marks]

• **[Boxed Result: Maximum Z = 16 at point (0, 4)]**

4. Solve graphically: Maximize Z = 4x + y subject to x + y <= 50, 3x + y <= 90, x >= 0, y >= 0. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Lines & Corner Points]** Maximize Z = 4x + y subject to x + y ≤ 50, 3x + y ≤ 90, x ≥ 0, y ≥ 0. Line 1: x + y = 50 through (50, 0) and (0, 50). Line 2: 3x + y = 90 through (30, 0) and (0, 90). Intersection: (3x + y) - (x + y) = 90 - 50 ⇒ 2x = 40 ⇒ x = 20; y = 30. Corner points: (0, 0), (30, 0), (20, 30), (0, 50). [1.5 Marks]

• **[Evaluations]** (0, 0): Z = 0. (30, 0): Z = 4(30) + 0 = 120 (Maximum). (20, 30): Z = 4(20) + 30 = 80 + 30 = 110. (0, 50): Z = 0 + 50 = 50. [1.5 Marks]

• **[Boxed Result: Maximum Z = 120 at point (30, 0)]**

5. Solve graphically: Minimize Z = -3x + 4y subject to x + 2y <= 8, 3x + 2y <= 12, x >= 0, y >= 0. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Lines & Corner Points]** Minimize Z = -3x + 4y subject to x + 2y ≤ 8, 3x + 2y ≤ 12, x ≥ 0, y ≥ 0. Line 1: x + 2y = 8 meets axes at (8, 0) and (0, 4). Line 2: 3x + 2y = 12 meets axes at (4, 0) and (0, 6). Intersection: Subtracting equations gives 2x = 4 ⇒ x = 2; then 2 + 2y = 8 ⇒ y = 3. Point is (2, 3). Corner points of bounded feasible polygon: (0, 0), (4, 0), (2, 3), (0, 4). [2 Marks]

• **[Evaluation Table]** (0, 0): Z = 0. (4, 0): Z = -3(4) + 4(0) = -12 (Minimum). (2, 3): Z = -3(2) + 4(3) = -6 + 12 = 6. (0, 4): Z = -3(0) + 4(4) = 16. [2 Marks]

• **[Boxed Result: Minimum Z = -12 at point (4, 0)]**

6. Solve graphically: Maximize Z = 5x + 3y subject to 3x + 5y <= 15, 5x + 2y <= 10, x >= 0, y >= 0. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Lines & Feasible Region]** Maximize Z = 5x + 3y subject to 3x + 5y ≤ 15, 5x + 2y ≤ 10, x ≥ 0, y ≥ 0. Line 1: 3x + 5y = 15 passes through (5, 0) and (0, 3). Line 2: 5x + 2y = 10 passes through (2, 0) and (0, 5). Intersection: 3x + 5y = 15 and 5x + 2y = 10 ⇒ x = 20 over 19, y = 45 over 19. Corner points: (0, 0), (2, 0), (20 over 19, 45 over 19), (0, 3). [2 Marks]

• **[Z Evaluations]** (0, 0): Z = 0. (2, 0): Z = 5(2) + 0 = 10. (20 over 19, 45 over 19): Z = 5(20 over 19) + 3(45 over 19) = (100 + 135) over 19 = 235 over 19 ≈ 12.37 (Maximum). (0, 3): Z = 0 + 3(3) = 9. [2 Marks]

• **[Boxed Result: Maximum Z = 235 over 19 at point (20 over 19, 45 over 19)]**

7. Solve graphically: Minimize Z = 3x + 5y subject to x + 3y >= 3, x + y >= 2, x >= 0, y >= 0. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Lines & Feasible Region]** Minimize Z = 3x + 5y subject to x + 3y ≥ 3, x + y ≥ 2, x ≥ 0, y ≥ 0. Line 1: x + 3y = 3 through (3, 0) and (0, 1). Line 2: x + y = 2 through (2, 0) and (0, 2). Intersection: Subtracting: 2y = 1 ⇒ y = 1 over 2; x = 3 over 2. Point (3 over 2, 1 over 2). Unbounded feasible region corner points: (0, 2), (3 over 2, 1 over 2), (3, 0). [2 Marks]

• **[Evaluation & Unbounded Check]** (0, 2): Z = 3(0) + 5(2) = 10. (3 over 2, 1 over 2): Z = 3(3 over 2) + 5(1 over 2) = 14 over 2 = 7 (Minimum). (3, 0): Z = 3(3) + 0 = 9. Open half plane 3x + 5y < 7 has no common points with feasible region. Minimum Z = 7 at (3 over 2, 1 over 2). [2 Marks]

• **[Boxed Result: Minimum Z = 7 at point (3 over 2, 1 over 2)]**